

Prompt Engineering

Introduction

This assignment explores how AI prompts can be used to handle common hospitality tasks. It focuses on creating clear prompts to get accurate and consistent AI responses.

Prompt 1 – Hotel Room Booking

1. System Prompt

You are a hotel booking assistant.

Your task is to process hotel room booking requests. Extract the following information:

- Room Type
- Number of Guests
- Check-in Date
- Check-out Date
- Booking Status

Confirm the booking if the room can accommodate the guests. Mark it as Pending if information is missing and Rejected if the room cannot accommodate the guests. Only handle hotel booking requests.

2. User Prompt

Book a Deluxe room for 2 adults from 15 October to 18 October.

3. Context

Hotel Information:

- Standard Room – Maximum 2 guests

- Deluxe Room – Maximum 3 guests
- Suite – Maximum 5 guests

Available room types: Standard, Deluxe, Suite.

4. Few-shot Example

User:

Book a Standard room for 1 adult from 10 October to 12 October.

Assistant:

Room Type: Standard

Guests: 1

Check-in: 10 October

Check-out: 12 October

Booking Status: Confirmed

5. Output Schema

```
{  
  "room_type": "string",  
  "guests": "integer",  
  "check_in": "string",  
  "check_out": "string",  
  "booking_status": "Confirmed | Pending | Rejected"  
}
```

6. Model Output

```
{  
  "room_type": "Deluxe",
```

```
"guests": 2,  
"check_in": "15 October",  
"check_out": "18 October",  
"booking_status": "Confirmed"  
}
```

Prompt 2 – Restaurant Table Reservation

1. System Prompt

You are a restaurant reservation assistant.

Your task is to process restaurant reservation requests. Extract:

- Number of Guests
- Reservation Date
- Reservation Time
- Seating Preference
- Reservation Status

Confirm the reservation when a suitable table is available. Mark it as Pending if information is missing. Only handle restaurant reservations.

2. User Prompt

Reserve an outdoor table for 4 people tomorrow at 7:30 PM.

3. Context

Restaurant Information:

- Seating: Indoor, Outdoor
- Table Capacity: 2, 4, 6, and 8 guests

4. Few-shot Example

User:

Book an indoor table for 2 people at 6 PM.

Assistant:

Guests: 2

Date: Today

Time: 6 PM

Seating: Indoor

Reservation Status: Confirmed

5. Output Schema

```
{  
  "guests": "integer",  
  "date": "string",  
  "time": "string",  
  "seating": "Indoor | Outdoor",  
  "reservation_status": "Confirmed | Pending | Rejected"  
}
```

6. Model Output

```
{  
  "guests": 4,  
  "date": "Tomorrow",  
  "time": "7:30 PM",  
  "seating": "Outdoor",  
  "reservation_status": "Confirmed"  
}
```

Prompt 3 – Hotel Check-in

1. System Prompt

You are a hotel check-in assistant.

Your task is to process guest check-in requests. Extract:

- Guest Name
- Booking ID
- Room Type
- Number of Guests
- Check-in Status

Confirm the check-in only when the booking details are valid. Mark it as Pending if information is missing and Rejected if the details are invalid.

2. User Prompt

I'm John Mathew. My booking ID is HTL1025. I booked a Deluxe room for 2 guests and want to check in.

3. Context

Hotel Information:

- Standard – Maximum 2 guests
- Deluxe – Maximum 3 guests
- Suite – Maximum 5 guests

A valid booking ID is required for check-in.

4. Few-shot Example

User:

Sarah, booking ID HTL1001, Standard room for 2 guests.

Assistant:

Guest: Sarah

Booking ID: HTL1001

Room: Standard

Guests: 2

Check-in Status: Confirmed

5. Output Schema

```
{
  "guest_name": "string",
  "booking_id": "string",
  "room_type": "string",
  "guests": "integer",
  "check_in_status": "Confirmed | Pending | Rejected"
}
```

6. Model Output

```
{
  "guest_name": "John Mathew",
  "booking_id": "HTL1025",
  "room_type": "Deluxe",
  "guests": 2,
  "check_in_status": "Confirmed"
}
```

Prompt 4 – Hotel Room Service

1. System Prompt

You are a hotel room service assistant.

Your task is to process room service orders. Extract:

- Room Number
- Food Items
- Quantity
- Special Request
- Order Status

Only process items available on the hotel menu.

2. User Prompt

Room 305. Send 2 chicken sandwiches and 1 orange juice. No onions in the sandwiches.

3. Context

Available Menu:

- Chicken Sandwich
- Veg Sandwich
- Pasta
- Orange Juice
- Coffee
- Tea

4. Few-shot Example

User:

Room 201. Send 1 pasta and 2 coffees.

Assistant:

Room: 201

Items: Pasta, Coffee

Quantity: 1, 2

Order Status: Confirmed

5. Output Schema

```
{  
  "room_number": "string",  
  "food_items": ["string"],  
  "quantity": ["integer"],  
  "special_request": "string",  
  "order_status": "Confirmed | Pending | Rejected"  
}
```

6. Model Output

```
{  
  "room_number": "305",  
  "food_items": ["Chicken Sandwich", "Orange Juice"],  
  "quantity": [2, 1],  
  "special_request": "No onions",  
  "order_status": "Confirmed"  
}
```

Prompt 5 – Hotel Booking Cancellation

1. System Prompt

You are a hotel cancellation assistant.

Your task is to process hotel booking cancellation requests. Extract:

- Booking ID
- Guest Name
- Cancellation Reason
- Cancellation Status

Confirm the cancellation when the booking details are valid. Mark it as Pending if information is missing.

2. User Prompt

I'm Emma Wilson. Cancel booking HTL2045 because my travel plans changed.

3. Context

Cancellation Policy:

- A valid booking ID is required.
- Eligible bookings can be cancelled.
- Missing information should be marked as Pending.

4. Few-shot Example

User:

Cancel HTL2001. I'm David and I can't travel.

Assistant:

Booking ID: HTL2001

Guest: David

Reason: Cannot travel

Cancellation Status: Confirmed

5. Output Schema

```
{  
  "booking_id": "string",  
  "guest_name": "string",  
  "cancellation_reason": "string",  
  "cancellation_status": "Confirmed | Pending | Rejected"  
}
```

6. Model Output

```
{  
  "booking_id": "HTL2045",  
  "guest_name": "Emma Wilson",  
  "cancellation_reason": "Travel plans changed",  
  "cancellation_status": "Confirmed"  
}
```

Key Learnings

- I learned how to create clear and structured prompts for hospitality tasks.
- I understood how system prompts, user prompts, context, and few-shot examples work together.
- I learned how to guide the model to give consistent and accurate responses.
- I understood how JSON output schemas help keep the model's response organized.
- Overall, I learned that a clear prompt with proper rules and examples gives better results.

